

DAY 2 — DATA UNDERSTANDING

Data Dictionary & Data Quality Assessment

Project: E-Commerce Sales & Customer Analytics System

Dataset: Sample - Superstore.csv

1. Dataset Overview

Item	Result
Total Records	9,994
Total Columns	21
Order Date Range	03 Jan 2014 – 30 Dec 2017
Unique Customers	793
Unique Products	1,862
Categories	3
Sub-Categories	17
Regions	4
Segments	3
States	49
Cities	531

2. Data Dictionary

Column	Data Type	Business Meaning
Row ID	Integer	Unique ID for each row.
Order ID	String	Unique order identifier.
Order Date	Date	Date when the order was placed.
Ship Date	Date	Date when the product was shipped.
Ship Mode	Category	Shipping mode specified by the customer.
Customer ID	String	Unique ID used to identify each customer.
Customer Name	String	Name of the customer.
Segment	Category	Customer segment.
Country	Category	Country of residence of the customer.
City	Category	City of residence of the customer.
State	Category	State of residence of the customer.
Postal Code	Integer	Postal code of the customer.
Region	Category	Region associated with the customer/order.
Product ID	String	Unique ID of the product.
Category	Category	Category of the product ordered.
Sub-Category	Category	Sub-category of the product ordered.

Column	Data Type	Business Meaning
Product Name	String	Name of the product.
Sales	Decimal	Sales/revenue amount for the record.
Quantity	Integer	Quantity of the product ordered.
Discount	Decimal	Discount provided.
Profit	Decimal	Profit or loss incurred.

3. Data Quality Assessment

Check	Result	Observation
Missing Values	0 in all columns	No missing values detected.
Duplicate Rows	0	No complete duplicate rows detected.
Order Date Parsing	0 failures	All Order Date values parsed successfully.
Ship Date Parsing	0 failures	All Ship Date values parsed successfully.
Ship Date < Order Date	0	No shipping dates occur before order dates.
Sales ≤ 0	0	No zero/negative sales records detected.
Quantity ≤ 0	0	No zero/negative quantity records detected.
Discount outside 0–1	0	All discount values fall within 0 to 1.
Postal Code = 0	0	No zero postal-code records detected.

4. Categorical Value Summary

Column	Observed Values
Ship Mode	First Class, Same Day, Second Class, Standard Class
Segment	Consumer, Corporate, Home Office
Country	United States
Region	Central, East, South, West
Category	Furniture, Office Supplies, Technology

5. Numeric Data Summary

Statistic	Sales	Quantity	Discount	Profit
Count	9,994.00	9,994.00	9,994.00	9,994.00
Mean	229.86	3.79	0.16	28.66
Median	54.49	3.00	0.20	8.67
Min	0.44	1.00	0.00	-6,599.98
Max	22,638.48	14.00	0.80	8,399.98

6. Potential Outliers for Further Review

An IQR-based screening identified potentially unusual observations in several numeric fields. These values are not automatically treated as errors and are not removed during Day 2. They should be reviewed during the Day 3 cleaning stage before any decision is made.

Column	Potential IQR Outliers
Sales	1,167
Quantity	170
Discount	856
Profit	1,881

7. Day 2 Conclusion

The dataset contains 9,994 records and 21 columns covering orders, customers, products, locations, sales, discounts, quantities, and profit. The initial assessment found no missing values or complete duplicate rows, and the basic validity checks identified no obvious invalid values. Potential outliers have been flagged for further review during data cleaning. The dataset is therefore ready for the next stage of the project: Excel data cleaning and preparation.